PROBLEMS

The symbols to the left of some of the problems (or their parts) have the following meaning:

D: The corresponding demonstration example just listed in Learning Aids may be helpful.
I: We suggest that you use the corresponding interactive procedure just listed (the printout records your work).
C: Use the computer with any of the software options available to you (or as instructed by your instructor) to solve the problem.

An asterisk on the problem number indicates that at least a partial answer is given in the back of the book.

12.1-1. Reconsider the California Manufacturing Co. example presented in Sec. 12.1. The mayor of San Diego now has contacted the company's president to try to persuade him to build a factory and perhaps a warehouse in that city. With the tax incentives being offered the company, the president's staff estimates that the net present value of building a factory in San Diego would be $7 million and the amount of capital required to do this would be $4 million. The net present value of building a warehouse there would be $5 million and the capital required would be $3 million. (This option would be considered only if a factory also is being built there.)

The company president now wants the previous OR study revised to incorporate these new alternatives into the overall problem. The objective still is to find the feasible combination of investments that maximizes the total net present value, given that the amount of capital available for these investments is $10 million.
(a) Formulate a BIP model for this problem.
(b) Display this model on an Excel spreadsheet.
C **(c)** Use the computer to solve this model.

12.1-2* A young couple, Eve and Steven, want to divide their main household chores (marketing, cooking, dishwashing, and laundering) between them so that each has two tasks but the total time they spend on household duties is kept to a minimum. Their efficiencies on these tasks differ, where the time each would need to perform the task is given by the following table:

	Time Needed per Week			
	Marketing	**Cooking**	**Dishwashing**	**Laundry**
Eve	4.5 hours	7.8 hours	3.6 hours	2.9 hours
Steven	4.9 hours	7.2 hours	4.3 hours	3.1 hours

(a) Formulate a BIP model for this problem.
(b) Display this model on an Excel spreadsheet.
C **(c)** Use the computer to solve this model.

12.1-3. A real estate development firm, Peterson and Johnson, is considering five possible development projects. The following table shows the estimated long-run profit (net present value) that each project would generate, as well as the amount of investment required to undertake the project, in units of millions of dollars.

	Development Project				
	1	**2**	**3**	**4**	**5**
Estimated profit	1	1.8	1.6	0.8	1.4
Capital required	6	12	10	4	8

The owners of the firm, Dave Peterson and Ron Johnson, have raised $20 million of investment capital for these projects. Dave and Ron now want to select the combination of projects that will maximize their total estimated long-run profit (net present value) without investing more that $20 million.
(a) Formulate a BIP model for this problem.
(b) Display this model on an Excel spreadsheet.
C **(c)** Use the computer to solve this model.

12.1-4. The board of directors of General Wheels Co. is considering six large capital investments. Each investment can be made only once. These investments differ in the estimated long-run profit (net present value) that they will generate as well as in the amount of capital required, as shown by the following table (in units of millions of dollars):

	Investment Opportunity					
	1	**2**	**3**	**4**	**5**	**6**
Estimated profit	15	12	16	18	9	11
Capital required	38	33	39	45	23	27

The total amount of capital available for these investments is $100 million. Investment opportunities 1 and 2 are mutually exclusive, and so are 3 and 4. Furthermore, neither 3 nor 4 can be undertaken unless one of the first two opportunities is undertaken. There are no such restrictions on investment opportunities 5 and 6. The objective is to select the combination of capital investments that will maximize the total estimated long-run profit (net present value).
(a) Formulate a BIP model for this problem.
C **(b)** Use the computer to solve this model.

12.1-5. Reconsider Prob. 9.3-4, where a swim team coach needs to assign swimmers to the different legs of a 200-yard medley relay team. Formulate a BIP model for this problem. Identify the groups of mutually exclusive alternatives in this formulation.

12.1-6. Vincent Cardoza is the owner and manager of a machine shop that does custom order work. This Wednesday afternoon, he has received calls from two customers who would like to place rush orders. One is a trailer hitch company which would like some custom-made heavy-duty tow bars. The other is a mini-car-carrier company which needs some customized stabilizer bars. Both customers would like as many as possible by the end of the week (two working days). Since both products would require the use of the same two machines, Vincent needs to decide and inform

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the customers this afternoon about how many of each product he will agree to make over the next two days.

Each tow bar requires 3.2 hours on machine 1 and 2 hours on machine 2. Each stabilizer bar requires 2.4 hours on machine 1 and 3 hours on machine 2. Machine 1 will be available for 16 hours over the next two days and machine 2 will be available for 15 hours. The profit for each tow bar produced would be \$130 and the profit for each stabilizer bar produced would be \$150.

Vincent now wants to determine the mix of these production quantities that will maximize the total profit.

(a) Formulate an IP model for this problem.

(b) Use a graphical approach to solve this model.

C **(c)** Use the computer to solve the model.

12.1-7. Reconsider Prob. 9.2-21 involving a contractor (Susan Meyer) who needs to arrange for hauling gravel from two pits to three building sites.

Susan now needs to hire the trucks (and their drivers) to do the hauling. Each truck can only be used to haul gravel from a single pit to a single site. In addition to the hauling and gravel costs specified in Prob. 9.2-21, there now is a fixed cost of \$50 associated with hiring each truck. A truck can haul 5 tons, but it is not required to go full. For each combination of pit and site, there are now two decisions to be made: the number of trucks to be used and the amount of gravel to be hauled.

(a) Formulate an MIP model for this problem.

C **(b)** Use the computer to solve this model.

12.2-1. Read the referenced article that fully describes the OR study summarized in the first application vignette presented in Sec. 12.2. Briefly describe how integer programming was applied in this study. Then list the various financial and nonfinancial benefits that resulted from this study.

12.2-2. Select one of the actual applications of BIP by a company or governmental agency mentioned in Sec. 12.2. Read the article describing the application in the referenced issue of *Interfaces.* Write a two-page summary of the application and its benefits.

12.2-3. Select three of the actual applications of BIP by a company or governmental agency mentioned in Sec. 12.2. Read the articles describing the applications in the referenced issues of *Interfaces.* For each one, write a one-page summary of the application and its benefits.

12.2-4. Follow the instructions of Prob. 12.2-1 for the second application vignette presented in Sec. 12.2.

12.3-1.* The Research and Development Division of the Progressive Company has been developing four possible new product lines. Management must now make a decision as to which of these four products actually will be produced and at what levels. Therefore, an operations research study has been requested to find the most profitable product mix.

A substantial cost is associated with beginning the production of any product, as given in the first row of the following table. Management's objective is to find the product mix that maximizes the total profit (total net revenue minus start-up costs).

	Product			
	1	2	3	4
Start-up cost	\$50,000	\$40,000	\$70,000	\$60,000
Marginal revenue	\$70	\$60	\$90	\$80

Let the continuous decision variables x_1, x_2, x_3, and x_4 be the production levels of products 1, 2, 3, and 4, respectively. Management has imposed the following policy constraints on these variables:

1. No more than two of the products can be produced.
2. Either product 3 or 4 can be produced only if either product 1 or 2 is produced.
3. Either $5x_1 + 3x_2 + 6x_3 + 4x_4 \leq 6{,}000$
   or $4x_1 + 6x_2 + 3x_3 + 5x_4 \leq 6{,}000.$

(a) Introduce auxiliary binary variables to formulate a mixed BIP model for this problem.

C **(b)** Use the computer to solve this model.

12.3-2. Suppose that a mathematical model fits linear programming except for the restriction that $|x_1 - x_2| = 0$, or 3, or 6. Show how to reformulate this restriction to fit an MIP model.

12.3-3. Suppose that a mathematical model fits linear programming except for the restrictions that

1. At least one of the following two inequalities holds:

$$3x_1 - x_2 - x_3 + x_4 \leq 12$$
$$x_1 + x_2 + x_3 + x_4 \leq 15.$$

2. At least two of the following three inequalities holds:

$$2x_1 + 5x_2 - x_3 + x_4 \leq 30$$
$$-x_1 + 3x_2 + 5x_3 + x_4 \leq 40$$
$$3x_1 - x_2 + 3x_3 - x_4 \leq 60.$$

Show how to reformulate these restrictions to fit an MIP model.

12.3-4. The Toys-R-4-U Company has developed two new toys for possible inclusion in its product line for the upcoming Christmas season. Setting up the production facilities to begin production would cost \$50,000 for toy 1 and \$80,000 for toy 2. Once these costs are covered, the toys would generate a unit profit of \$10 for toy 1 and \$15 for toy 2.

The company has two factories that are capable of producing these toys. However, to avoid doubling the start-up costs, just one factory would be used, where the choice would be based on maximizing profit. For administrative reasons, the same factory would be used for both new toys if both are produced.

Toy 1 can be produced at the rate of 50 per hour in factory 1 and 40 per hour in factory 2. Toy 2 can be produced at the rate of 40 per hour in factory 1 and 25 per hour in factory 2. Factories 1 and 2, respectively, have 500 hours and 700 hours of production time available before Christmas that could be used to produce these toys.

It is not known whether these two toys would be continued after Christmas. Therefore, the problem is to determine how many

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units (if any) of each new toy should be produced before Christmas to maximize the total profit.

(a) Formulate an MIP model for this problem.

C **(b)** Use the computer to solve this model.

12.3-5.* Northeastern Airlines is considering the purchase of new long-, medium-, and short-range jet passenger airplanes. The purchase price would be \$67 million for each long-range plane, \$50 million for each medium-range plane, and \$35 million for each short-range plane. The board of directors has authorized a maximum commitment of \$1.5 billion for these purchases. Regardless of which airplanes are purchased, air travel of all distances is expected to be sufficiently large that these planes would be utilized at essentially maximum capacity. It is estimated that the net annual profit (after capital recovery costs are subtracted) would be \$4.2 million per long-range plane, \$3 million per medium-range plane, and \$2.3 million per short-range plane.

It is predicted that enough trained pilots will be available to the company to crew 30 new airplanes. If only short-range planes were purchased, the maintenance facilities would be able to handle 40 new planes. However, each medium-range plane is equivalent to $1\frac{1}{3}$ short-range planes, and each long-range plane is equivalent to $1\frac{2}{3}$ short-range planes in terms of their use of the maintenance facilities.

The information given here was obtained by a preliminary analysis of the problem. A more detailed analysis will be conducted subsequently. However, using the preceding data as a first approximation, management wishes to know how many planes of each type should be purchased to maximize profit.

(a) Formulate an IP model for this problem.

C **(b)** Use the computer to solve this problem.

(c) Use a binary representation of the variables to reformulate the IP model in part (*a*) as a BIP problem.

C **(d)** Use the computer to solve the BIP model formulated in part (*c*). Then use this optimal solution to identify an optimal solution for the IP model formulated in part (*a*).

12.3-6. Consider the two-variable IP example discussed in Sec. 12.5 and illustrated in Fig. 12.3.

(a) Use a binary representation of the variables to reformulate this model as a BIP problem.

C **(b)** Use the computer to solve this BIP problem. Then use this optimal solution to identify an optimal solution for the original IP model.

12.3-7. The Fly-Right Airplane Company builds small jet airplanes to sell to corporations for the use of their executives. To meet the needs of these executives, the company's customers sometimes order a custom design of the airplanes being purchased. When this occurs, a substantial start-up cost is incurred to initiate the production of these airplanes.

Fly-Right has recently received purchase requests from three customers with short deadlines. However, because the company's production facilities already are almost completely tied up filling previous orders, it will not be able to accept all three orders. Therefore, a decision now needs to be made on the number of airplanes the company will agree to produce (if any) for each of the three customers.

The relevant data are given in the next table. The first row gives the start-up cost required to initiate the production of the airplanes for each customer. Once production is under way, the marginal net revenue (which is the purchase price minus the marginal production cost) from each airplane produced is shown in the second row. The third row gives the percentage of the available production capacity that would be used for each airplane produced. The last row indicates the maximum number of airplanes requested by each customer (but less will be accepted).

	Customer		
	1	**2**	**3**
Start-up cost	\$3 million	\$2 million	0
Marginal net revenue	\$2 million	\$3 million	\$0.8 million
Capacity used per plane	20%	40%	20%
Maximum order	3 planes	2 planes	5 planes

Fly-Right now wants to determine how many airplanes to produce for each customer (if any) to maximize the company's total profit (total net revenue minus start-up costs).

(a) Formulate a model with both integer variables and binary variables for this problem.

C **(b)** Use the computer to solve this model.

12.4-1. Reconsider the Fly-Right Airplane Co. problem introduced in Prob. 12.3-7. A more detailed analysis of the various cost and revenue factors now has revealed that the potential profit from producing airplanes for each customer cannot be expressed simply in terms of a *start-up cost* and a fixed *marginal net revenue* per airplane produced. Instead, the profits are given by the following table.

Airplanes Produced	**Profit from Customer**		
	1	**2**	**3**
0	0	0	0
1	−\$1 million	\$1 million	\$1 million
2	\$2 million	\$5 million	\$3 million
3	\$4 million		\$5 million
4			\$6 million
5			\$7 million

(a) Formulate a BIP model for this problem that includes constraints for *mutually exclusive alternatives.*

C **(b)** Use the computer to solve the model formulated in part (*a*). Then use this optimal solution to identify the optimal number of airplanes to produce for each customer.

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(c) Formulate another BIP model for this model that includes constraints for *contingent decisions.*

C **(d)** Repeat part (*b*) for the model formulated in part (*c*).

12.4-2. Reconsider the Wyndor Glass Co. problem presented in Sec. 3.1. Management now has decided that only one of the two new products should be produced, and the choice is to be made on the basis of maximizing profit. Introduce *auxiliary binary variables* to formulate an MIP model for this new version of the problem.

12.4-3.* Reconsider Prob. 3.1-11, where the management of the Omega Manufacturing Company is considering devoting excess production capacity to one or more of three products. (See the Partial Answers to Selected Problems in the back of the book for additional information about this problem.) Management now has decided to add the restriction that no more than two of the three prospective products should be produced.

(a) Introduce *auxiliary binary variables* to formulate an MIP model for this new version of the problem.

C **(b)** Use the computer to solve this model.

12.4-4. Consider the following integer nonlinear programming problem:

$$\text{Maximize} \qquad Z = 4x_1^2 - x_1^3 + 10x_2^2 - x_2^4,$$

subject to

$$x_1 + x_2 \leq 3$$

and

$$x_1 \geq 0, \qquad x_2 \geq 0$$
$$x_1 \text{ and } x_2 \text{ are integers.}$$

This problem can be reformulated in two different ways as an equivalent pure BIP problem (with a linear objective function) with six binary variables (y_{1j} and y_{2j} for $j = 1, 2, 3$), depending on the interpretation given the binary variables.

(a) Formulate a BIP model for this problem where the binary variables have the interpretation,

$$y_{ij} = \begin{cases} 1 & \text{if } x_i = j \\ 0 & \text{otherwise.} \end{cases}$$

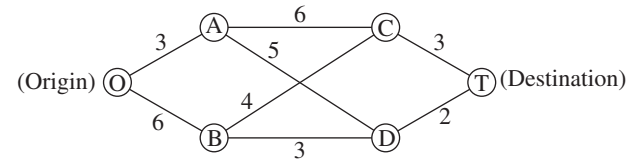
C **(b)** Use the computer to solve the model formulated in part (*a*), and thereby identify an optimal solution for (x_1, x_2) for the original problem.

(c) Formulate a BIP model for this problem where the binary variables have the interpretation,

$$y_{ij} = \begin{cases} 1 & \text{if } x_i \geq j \\ 0 & \text{otherwise.} \end{cases}$$

C **(d)** Use the computer to solve the model formulated in part (*c*), and thereby identify an optimal solution for (x_1, x_2) for the original problem.

12.4-5.* Consider the following special type of *shortest-path problem* (see Sec. 10.3) where the nodes are in columns and the only paths considered always move forward one column at a time.



The numbers along the links represent distances, and the objective is to find the shortest path from the origin to the destination.

This problem also can be formulated as a BIP model involving both mutually exclusive alternatives and contingent decisions.

(a) Formulate this model. Identify the constraints that are for mutually exclusive alternatives and that are for contingent decisions.

C **(b)** Use the computer to solve this problem.

12.4-6. Speedy Delivery provides two-day delivery service of large parcels across the United States. Each morning at each collection center, the parcels that have arrived overnight are loaded onto several trucks for delivery throughout the area. Since the competitive battlefield in this business is speed of delivery, the parcels are divided among the trucks according to their geographical destinations to minimize the average time needed to make the deliveries.

On this particular morning, the dispatcher for the Blue River Valley Collection Center, Sharon Lofton, is hard at work. Her three drivers will be arriving in less than an hour to make the day's deliveries. There are nine parcels to be delivered, all at locations many miles apart. As usual, Sharon has loaded these locations into her computer. She is using her company's special software package, a decision support system called Dispatcher. The first thing Dispatcher does is use these locations to generate a considerable number of attractive possible routes for the individual delivery trucks. These routes are shown in the following table (where the numbers in each column indicate the order of the deliveries), along with the estimated time required to traverse the route.

	Attractive Possible Route									
Delivery Location	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
A	1				1				1	
B		2		1		2			2	2
C			3	3			3		3	
D	2					1		1		
E			2	2		3				
F		1			2					
G	3						1	2		3
H			1		3					1
I		3		4			2			
Time (in hours)	6	4	7	5	4	6	5	3	7	6

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Dispatcher is an interactive system that shows these routes to Sharon for her approval or modification. (For example, the computer may not know that flooding has made a particular route infeasible.) After Sharon approves these routes as attractive possibilities with reasonable time estimates, Dispatcher next formulates and solves a BIP model for selecting three routes that minimize their total time while including each delivery location on exactly one route. This morning, Sharon does approve all the routes.

(a) Formulate this BIP model.

C **(b)** Use the computer to solve this model.

12.4-7. An increasing number of Americans are moving to a warmer climate when they retire. To take advantage of this trend, Sunny Skies Unlimited is undertaking a major real estate development project. The project is to develop a completely new retirement community (to be called Pilgrim Haven) that will cover several square miles. One of the decisions to be made is where to locate the two fire stations that have been allocated to the community. For planning purposes, Pilgrim Haven has been divided into five tracts, with no more than one fire station to be located in any given tract. Each station is to respond to *all* the fires that occur in the tract in which it is located as well as in the other tracts that are assigned to this station. Thus, the decisions to be made consist of (1) the tracts to receive a fire station and (2) the assignment of each of the other tracts to one of the fire stations. The objective is to minimize the overall average of the *response times* to fires.

The following table gives the average response time to a fire in each tract (the columns) if that tract is served by a station in a given tract (the rows). The bottom row gives the forecasted average number of fires that will occur in each of the tracts per day.

Assigned Station Located in Tract	Response Times (in minutes) Fire in Tract				
	1	2	3	4	5
1	5	12	30	20	15
2	20	4	15	10	25
3	15	20	6	15	12
4	25	15	25	4	10
5	10	25	15	12	5
Average frequency of fires	2 per day	1 per day	3 per day	1 per day	3 per day

Formulate a BIP model for this problem. Identify any constraints that correspond to mutually exclusive alternatives or contingent decisions.

12.4-8. Reconsider Prob. 12.4-7. The management of Sunny Skies Unlimited now has decided that the decision on the locations of the fire stations should be based mainly on costs.

The cost of locating a fire station in a tract is \$200,000 for tract 1, \$250,000 for tract 2, \$400,000 for tract 3, \$300,000 for tract 4, and \$500,000 for tract 5. Management's objective now is the following:

> Determine which tracts should receive a station to minimize the total cost of stations while ensuring that each tract has at least one station close enough to respond to a fire in no more than 15 minutes (on the average).

In contrast to the original problem, note that the total number of fire stations is no longer fixed. Furthermore, if a tract without a station has more than one station within 15 minutes, it is no longer necessary to assign this tract to just one of these stations.

(a) Formulate a complete pure BIP model with 5 binary variables for this problem.

(b) Is this a *set covering problem?* Explain, and identify the relevant sets.

C **(c)** Use the computer to solve the model formulated in part (*a*).

12.4-9. Suppose that a state sends R persons to the U.S. House of Representatives. There are D counties in the state ($D > R$), and the state legislature wants to group these counties into R distinct electoral districts, each of which sends a delegate to Congress. The total population of the state is P, and the legislature wants to form districts whose population approximates $p = P/R$. Suppose that the appropriate legislative committee studying the electoral districting problem generates a long list of N *candidates* to be districts ($N > R$). Each of these candidates contains contiguous counties and a total population p_j ($j = 1, 2, \ldots, N$) that is acceptably close to p. Define $c_j = |p_j - p|$. Each county i ($i = 1, 2, \ldots, D$) is included in at least one candidate and typically will be included in a considerable number of candidates (in order to provide many feasible ways of selecting a set of R candidates that includes each county exactly once). Define

$$a_{ij} = \begin{cases} 1 & \text{if county } i \text{ is included in candidate } j \\ 0 & \text{if not.} \end{cases}$$

Given the values of the c_j and the a_{ij}, the objective is to select R of these N possible districts such that each county is contained in a single district and such that the largest of the associated c_j is as small as possible.

Formulate a BIP model for this problem.

12.5-1. Read the referenced article that fully describes the OR study summarized in the application vignette presented in Sec. 12.5. Briefly describe how integer programming was applied in this study. Then list the various financial and nonfinancial benefits that resulted from this study.

12.5-2.* Consider the following IP problem:

$$\text{Maximize} \quad Z = 5x_1 + x_2,$$

subject to

$$\begin{aligned} -x_1 + 2x_2 &\leq 4 \\ x_1 - x_2 &\leq 1 \\ 4x_1 + x_2 &\leq 12 \end{aligned}$$

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and

$$x_1 \geq 0, \qquad x_2 \geq 0$$
$$x_1, x_2 \text{ are integers.}$$

(a) Solve this problem graphically.

(b) Solve the LP relaxation graphically. Round this solution to the *nearest* integer solution and check whether it is feasible. Then enumerate *all* the rounded solutions by rounding this solution for the LP relaxation in *all* possible ways (i.e., by rounding each noninteger value both up and down). For each rounded solution, check for feasibility and, if feasible, calculate Z. Are any of these feasible rounded solutions optimal for the IP problem?

12.5-3. Follow the instructions of Prob. 12.5-2 for the following IP problem:

$$\text{Maximize} \qquad Z = 220x_1 + 80x_2,$$

subject to

$$\begin{aligned} 5x_1 + 2x_2 &\leq 16 \\ 2x_1 - x_2 &\leq 4 \\ -x_1 + 2x_2 &\leq 4 \end{aligned}$$

and

$$x_1 \geq 0, \qquad x_2 \geq 0$$
$$x_1, x_2 \text{ are integers.}$$

12.5-4. Follow the instructions of Prob. 12.5-2 for the following BIP problem:

$$\text{Maximize} \qquad Z = 2x_1 + 5x_2,$$

subject to

$$\begin{aligned} 10x_1 + 30x_2 &\leq 30 \\ 95x_1 - 30x_2 &\leq 75 \end{aligned}$$

and

$$x_1, x_2 \text{ are binary.}$$

12.5-5. Follow the instructions of Prob. 12.5-2 for the following BIP problem:

$$\text{Maximize} \qquad Z = -5x_1 + 25x_2,$$

subject to

$$\begin{aligned} -3x_1 + 30x_2 &\leq 27 \\ 3x_1 + x_2 &\leq 4 \end{aligned}$$

and

$$x_1, x_2 \text{ are binary.}$$

12.5-6. Label each of the following statements as True or False, and then justify your answer by referring to specific statements in the chapter:

(a) Linear programming problems are generally considerably easier to solve than IP problems.

(b) For IP problems, the number of integer variables is generally more important in determining the computational difficulty than is the number of functional constraints.

(c) To solve an IP problem with an approximate procedure, one may apply the simplex method to the LP relaxation problem and then round each noninteger value to the nearest integer. The result will be a feasible but not necessarily optimal solution for the IP problem.

D,I **12.6-1.*** Use the BIP branch-and-bound algorithm presented in Sec. 12.6 to solve the following problem interactively:

$$\text{Maximize} \qquad Z = 2x_1 - x_2 + 5x_3 - 3x_4 + 4x_5,$$

subject to

$$\begin{aligned} 3x_1 - 2x_2 + 7x_3 - 5x_4 + 4x_5 &\leq 6 \\ x_1 - x_2 + 2x_3 - 4x_4 + 2x_5 &\leq 0 \end{aligned}$$

and

$$x_j \text{ is binary,} \qquad \text{for } j = 1, 2, \ldots, 5.$$

D,I **12.6-2.** Use the BIP branch-and-bound algorithm presented in Sec. 12.6 to solve the following problem interactively:

$$\text{Minimize} \qquad Z = 5x_1 + 6x_2 + 7x_3 + 8x_4 + 9x_5,$$

subject to

$$\begin{aligned} 3x_1 - x_2 + x_3 + x_4 - 2x_5 &\geq 2 \\ x_1 + 3x_2 - x_3 - 2x_4 + x_5 &\geq 0 \\ -x_1 - x_2 + 3x_3 + x_4 + x_5 &\geq 1 \end{aligned}$$

and

$$x_j \text{ is binary,} \qquad \text{for } j = 1, 2, \ldots, 5.$$

D,I **12.6-3.** Use the BIP branch-and-bound algorithm presented in Sec. 12.6 to solve the following problem interactively:

$$\text{Maximize} \qquad Z = 5x_1 + 5x_2 + 8x_3 - 2x_4 - 4x_5,$$

subject to

$$\begin{aligned} -3x_1 + 6x_2 - 7x_3 + 9x_4 + 9x_5 &\geq 10 \\ x_1 + 2x_2 \qquad - x_4 - 3x_5 &\leq 0 \end{aligned}$$

and

$$x_j \text{ is binary,} \qquad \text{for } j = 1, 2, \ldots, 5.$$

D,I **12.6-4.** Reconsider Prob. 12.3-6(*a*). Use the BIP branch-and-bound algorithm presented in Sec. 12.6 to solve this BIP model interactively.

D,I **12.6-5.** Reconsider Prob. 12.4-8(*a*). Use the BIP algorithm presented in Sec. 12.6 to solve this problem interactively.

12.6-6. Consider the following statements about any pure IP problem (in maximization form) and its LP relaxation. Label each of the statements as True or False, and then justify your answer:

(a) The feasible region for the LP relaxation is a subset of the feasible region for the IP problem.

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(b) If an optimal solution for the LP relaxation is an integer solution, then the optimal value of the objective function is the same for both problems.

(c) If a noninteger solution is feasible for the LP relaxation, then the nearest integer solution (rounding each variable to the nearest integer) is a feasible solution for the IP problem.

12.6-7.* Consider the assignment problem with the following cost table:

		Task				
		1	**2**	**3**	**4**	**5**
	1	39	65	69	66	57
	2	64	84	24	92	22
Assignee	3	49	50	61	31	45
	4	48	45	55	23	50
	5	59	34	30	34	18

(a) Design a branch-and-bound algorithm for solving such assignment problems by specifying how the branching, bounding, and fathoming steps would be performed. (*Hint:* For the assignees not yet assigned for the current subproblem, form the relaxation by deleting the constraints that each of these assignees must perform exactly one task.)

(b) Use this algorithm to solve this problem.

12.6-8. Five jobs need to be done on a certain machine. However, the setup time for each job depends upon which job immediately preceded it, as shown by the following table:

		Setup Time				
		Job				
		1	**2**	**3**	**4**	**5**
	None	4	5	8	9	4
	1	—	7	12	10	9
Immediately	2	6	—	10	14	11
Preceding Job	3	10	11	—	12	10
	4	7	8	15	—	7
	5	12	9	8	16	—

The objective is to schedule the *sequence* of jobs that minimizes the sum of the resulting setup times.

(a) Design a branch-and-bound algorithm for sequencing problems of this type by specifying how the branch, bound, and fathoming steps would be performed.

(b) Use this algorithm to solve this problem.

12.6-9.* Consider the following *nonlinear* BIP problem:

$$\text{Maximize} \qquad Z = 80x_1 + 60x_2 + 40x_3 + 20x_4 - (7x_1 + 5x_2 + 3x_3 + 2x_4)^2,$$

subject to

$$x_j \text{ is binary}, \qquad \text{for } j = 1, 2, 3, 4.$$

Given the value of the first k variables $x_1, \ldots, x_k$, where $k = 0$, 1, 2, or 3, an upper bound on the value of Z that can be achieved by the corresponding feasible solutions is

$$\sum_{j=1}^{k} c_j x_j - \left(\sum_{j=1}^{k} d_j x_j\right)^2 + \sum_{j=k+1}^{4} \max\left\{0,\ c_j - \left[\left(\sum_{i=1}^{k} d_i x_i + d_j\right)^2 - \left(\sum_{i=1}^{k} d_i x_i\right)^2\right]\right\},$$

where $c_1 = 80$, $c_2 = 60$, $c_3 = 40$, $c_4 = 20$, $d_1 = 7$, $d_2 = 5$, $d_3 = 3$, $d_4 = 2$. Use this bound to solve the problem by the branch-and-bound technique.

12.6-10. Consider the Lagrangian relaxation described near the end of Sec. 12.6.

(a) If $\mathbf{x}$ is a feasible solution for an MIP problem, show that $\mathbf{x}$ also must be a feasible solution for the corresponding Lagrangian relaxation.

(b) If $\mathbf{x}^*$ is an optimal solution for an MIP problem, with an objective function value of Z, show that $Z \leq Z_R^*$, where Z_R^* is the optimal objective function value for the corresponding Lagrangian relaxation.

12.7-1. Read the referenced article that fully describes the OR study summarized in the application vignette presented in Sec. 12.7. Briefly describe how integer programming was applied in this study. Then list the various financial and nonfinancial benefits that resulted from this study.

12.7-2.* Consider the following IP problem:

$$\text{Maximize} \qquad Z = -3x_1 + 5x_2,$$

subject to

$$5x_1 - 7x_2 \geq 3$$

and

$$x_j \leq 3$$
$$x_j \geq 0$$
$$x_j \text{ is integer}, \qquad \text{for } j = 1, 2.$$

(a) Solve this problem graphically.

(b) Use the MIP branch-and-bound algorithm presented in Sec. 12.7 to solve this problem by hand. For each subproblem, solve its LP relaxation *graphically*.

(c) Use the binary representation for integer variables to reformulate this problem as a BIP problem.

D,I **(d)** Use the BIP branch-and-bound algorithm presented in Sec. 12.6 to solve the problem as formulated in part (*c*) interactively.

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12.7-3. Follow the instructions of Prob. 12.7-2 for the following IP model:

$$\text{Minimize} \qquad Z = 2x_1 + 3x_2,$$

subject to

$$\begin{aligned} x_1 + x_2 &\geq 3 \\ x_1 + 3x_2 &\geq 6 \end{aligned}$$

and

$$x_1 \geq 0, \qquad x_2 \geq 0$$
$$x_1, x_2 \text{ are integers.}$$

12.7-4. Reconsider the IP model of Prob. 12.5-2.

(a) Use the MIP branch-and-bound algorithm presented in Sec. 12.7 to solve this problem by hand. For each subproblem, solve its LP relaxation *graphically.*

D,I **(b)** Now use the interactive procedure for this algorithm in your IOR Tutorial to solve this problem.

C **(c)** Check your answer by using an automatic procedure to solve the problem.

D,I **12.7-5.** Consider the IP example discussed in Sec. 12.5 and illustrated in Fig. 12.3. Use the MIP branch-and-bound algorithm presented in Sec. 12.7 to solve this problem interactively.

D,I **12.7-6.** Reconsider Prob. 12.3-5*a*. Use the MIP branch-and-bound algorithm presented in Sec. 12.7 to solve this IP problem interactively.

12.7-7. A machine shop makes two products. Each unit of the first product requires 3 hours on machine 1 and 2 hours on machine 2. Each unit of the second product requires 2 hours on machine 1 and 3 hours on machine 2. Machine 1 is available only 8 hours per day and machine 2 only 7 hours per day. The profit per unit sold is 16 for the first product and 10 for the second. The amount of each product produced per day must be an integral multiple of 0.25. The objective is to determine the mix of production quantities that will maximize profit.

(a) Formulate an IP model for this problem.

(b) Solve this model graphically.

(c) Use graphical analysis to apply the MIP branch-and-bound algorithm presented in Sec. 12.7 to solve this model.

D,I **(d)** Now use the interactive procedure for this algorithm in your IOR Tutorial to solve this model.

C **(e)** Check your answers in parts (*b*), (*c*), and (*d*) by using an automatic procedure to solve the model.

D,I **12.7-8.** Use the MIP branch-and-bound algorithm presented in Sec. 12.7 to solve the following MIP problem interactively:

$$\text{Maximize} \qquad Z = 5x_1 + 4x_2 + 4x_3 + 2x_4,$$

subject to

$$\begin{aligned} x_1 + 3x_2 + 2x_3 + x_4 &\leq 10 \\ 5x_1 + x_2 + 3x_3 + 2x_4 &\leq 15 \\ x_1 + x_2 + x_3 + x_4 &\leq 6 \end{aligned}$$

and

$$x_j \geq 0, \qquad \text{for } j = 1, 2, 3, 4$$
$$x_j \text{ is integer}, \qquad \text{for } j = 1, 2, 3.$$

D,I **12.7-9.** Use the MIP branch-and-bound algorithm presented in Sec. 12.7 to solve the following MIP problem interactively:

$$\text{Maximize} \qquad Z = 3x_1 + 4x_2 + 2x_3 + x_4 + 2x_5,$$

subject to

$$\begin{aligned} 2x_1 - x_2 + x_3 + x_4 + x_5 &\leq 3 \\ -x_1 + 3x_2 + x_3 - x_4 - 2x_5 &\leq 2 \\ 2x_1 + x_2 - x_3 + x_4 + 3x_5 &\leq 1 \end{aligned}$$

and

$$x_j \geq 0, \qquad \text{for } j = 1, 2, 3, 4, 5$$
$$x_j \text{ is binary}, \qquad \text{for } j = 1, 2, 3.$$

D,I **12.7-10.** Use the MIP branch-and-bound algorithm presented in Sec. 12.7 to solve the following MIP problem interactively:

$$\text{Minimize} \qquad Z = 5x_1 + x_2 + x_3 + 2x_4 + 3x_5,$$

subject to

$$\begin{aligned} x_2 - 5x_3 + x_4 + 2x_5 &\geq -2 \\ 5x_1 - x_2 \qquad\qquad + x_5 &\geq 7 \\ x_1 + x_2 + 6x_3 + x_4 \qquad\quad &\geq 4 \end{aligned}$$

and

$$x_j \geq 0, \qquad \text{for } j = 1, 2, 3, 4, 5$$
$$x_j \text{ is integer}, \qquad \text{for } j = 1, 2, 3.$$

12.8-1.* For each of the following constraints of pure BIP problems, use the constraint to fix as many variables as possible:

(a) $4x_1 + x_2 + 3x_3 + 2x_4 \leq 2$

(b) $4x_1 - x_2 + 3x_3 + 2x_4 \leq 2$

(c) $4x_1 - x_2 + 3x_3 + 2x_4 \geq 7$

12.8-2. For each of the following constraints of pure BIP problems, use the constraint to fix as many variables as possible:

(a) $20x_1 - 7x_2 + 5x_3 \leq 10$

(b) $10x_1 - 7x_2 + 5x_3 \geq 10$

(c) $10x_1 - 7x_2 + 5x_3 \leq -1$

12.8-3. Use the following set of constraints for the *same* pure BIP problem to fix as many variables as possible. Also identify the constraints which become redundant because of the fixed variables.

$$\begin{aligned} 3x_3 - x_5 + x_7 &\leq 1 \\ x_2 + x_4 + x_6 &\leq 1 \\ x_1 - 2x_5 + 2x_6 &\geq 2 \\ x_1 + x_2 - x_4 &\leq 0 \end{aligned}$$

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12.8-4. For each of the following constraints of pure BIP problems, identify which ones are made redundant by the binary constraints. Explain why each one is, or is not, redundant.
(a) $2x_1 + x_2 + 2x_3 \leq 5$
(b) $3x_1 - 4x_2 + 5x_3 \leq 5$
(c) $x_1 + x_2 + x_3 \geq 2$
(d) $3x_1 - x_2 - 2x_3 \geq -4$

12.8-5. In Sec. 12.8, at the end of the subsection on tightening constraints, we indicated that the constraint $4x_1 - 3x_2 + x_3 + 2x_4 \leq 5$ can be tightened to $2x_1 - 3x_2 + x_3 + 2x_4 \leq 3$ and then to $2x_1 - 2x_2 + x_3 + 2x_4 \leq 3$. Apply the procedure for tightening constraints to confirm these results.

12.8-6. Apply the procedure for *tightening constraints* to the following constraint for a pure BIP problem:

$$3x_1 - 2x_2 + x_3 \leq 3.$$

12.8-7. Apply the procedure for *tightening constraints* to the following constraint for a pure BIP problem:

$$x_1 - x_2 + 3x_3 + 4x_4 \geq 1.$$

12.8-8. Apply the procedure for *tightening constraints* to each of the following constraints for a pure BIP problem:
(a) $x_1 + 3x_2 - 4x_3 \leq 2.$
(b) $3x_1 - x_2 + 4x_3 \geq 1.$

12.8-9. In Sec. 12.8, a pure BIP example with the constraint, $2x_1 + 3x_2 \leq 4$, was used to illustrate the procedure for tightening constraints. Show that applying the procedure for generating cutting planes to this constraint yields the same new constraint, $x_1 + x_2 \leq 1$.

12.8-10. One of the constraints of a certain pure BIP problem is

$$x_1 + 3x_2 + 2x_3 + 4x_4 \leq 5.$$

Identify all the minimal covers for this constraint, and then give the corresponding cutting planes.

12.8-11. One of the constraints of a certain pure BIP problem is

$$3x_1 + 4x_2 + 2x_3 + 5x_4 \leq 7.$$

Identify all the minimal covers for this constraint, and then give the corresponding cutting planes.

12.8-12. Generate as many cutting planes as possible from the following constraint for a pure BIP problem:

$$3x_1 + 5x_2 + 4x_3 + 8x_4 \leq 10.$$

12.8-13. Generate as many cutting planes as possible from the following constraint for a pure BIP problem.

$$5x_1 + 3x_2 + 7x_3 + 4x_4 + 6x_5 \leq 9.$$

12.8-14. Consider the following BIP problem:

$$\text{Maximize} \quad Z = 2x_1 + 3x_2 + x_3 + 4x_4 + 3x_5 + 2x_6 + 2x_7 + x_8 + 3x_9,$$

subject to

$$\begin{aligned} 3x_2 + x_4 + x_5 &\geq 3 \\ x_1 + x_2 &\leq 1 \\ x_2 + x_4 - x_5 - x_6 &\leq -1 \\ x_2 + 2x_6 + 3x_7 + x_8 + 2x_9 &\geq 4 \\ -x_3 + 2x_5 + x_6 + 2x_7 - 2x_8 + x_9 &\leq 5 \end{aligned}$$

and

all x_j binary.

Develop the tightest possible formulation of this problem by using the techniques of automatic problem reprocessing (fixing variables, deleting redundant constraints, and tightening constraints). Then use this tightened formulation to determine an optimal solution by inspection.

12.9-1. Consider the following problem:

$$\text{Maximize } Z = 3x_1 + 2x_2 + 4x_3 + x_4,$$

subject to

$x_1 \in \{1, 3\}, \quad x_2 \in \{1, 2\}, \quad x_3 \in \{2, 3\}, \quad x_4 \in \{1, 2, 3, 4\},$
all these variables must have different values,
$x_1 + x_2 + x_3 + x_4 \leq 10.$

Use the techniques of constraint programming (domain reduction, constraint propagation, a search procedure, and enumeration) to identify all the feasible solutions and then to find an optimal solution. Show your work.

12.9-2. Consider the following problem:

$$\text{Maximize } Z = 5x_1 - x_1^2 + 8x_2 - x_2^2 + 10x_3 - x_3^2 + 15x_4 - x_4^2 + 20x_5 - x_5^2,$$

subject to

$x_1 \in \{3, 6, 12\}, x_2 \in \{3, 6\}, x_3 \in \{3, 6, 9, 12\},$
$x_4 \in \{6, 12\}, x_5 \in \{9, 12, 15, 18\},$
all these variables must have different values,
$x_1 + x_3 + x_4 \leq 25.$

Use the techniques of constraint programming (domain reduction, constraint propagation, a search procedure, and enumeration) to identify all the feasible solutions and then to find an optimal solution. Show your work.

12.9-3. Consider the following problem:

$$\text{Maximize } Z = 100x_1 - 3x_1^2 + 400x_2 - 5x_2^2 + 200x_3 - 4x_3^2 + 100x_4 - 2x_4^4,$$

subject to

$x_1 \in \{25, 30\}, x_2 \in \{20, 25, 30, 35, 40, 50\},$
$x_3 \in \{20, 25, 30\}, x_4 \in \{20, 25\},$
all these variables must have different values,
$x_2 + x_3 \leq 60,$
$x_1 + x_3 \leq 50.$

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